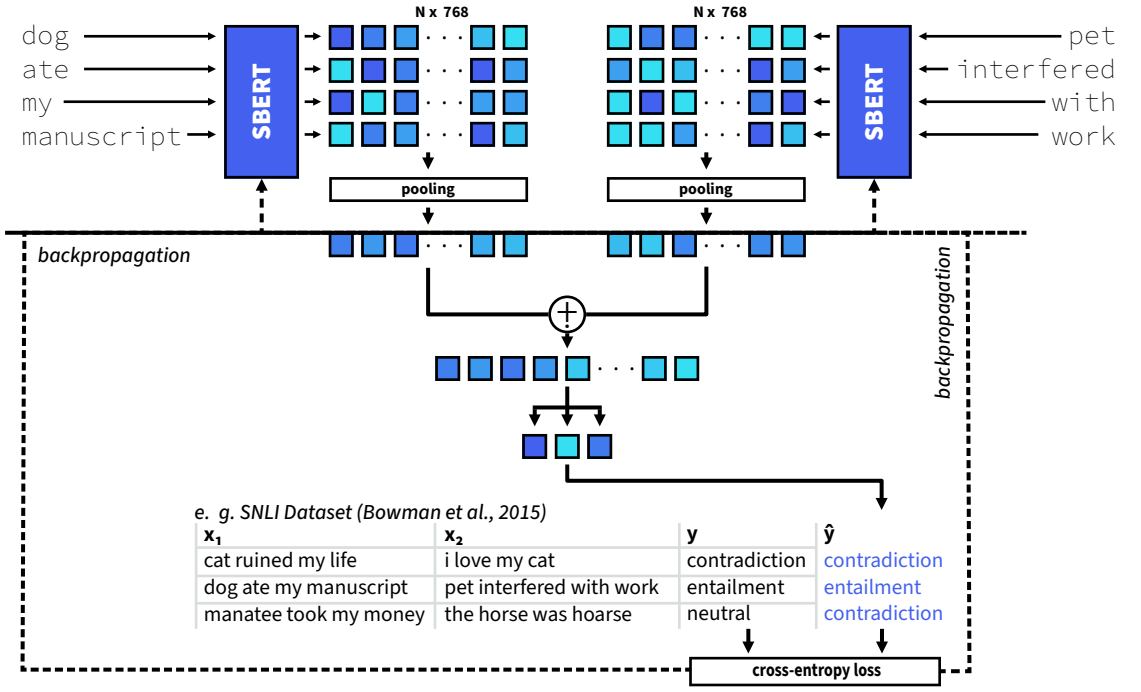
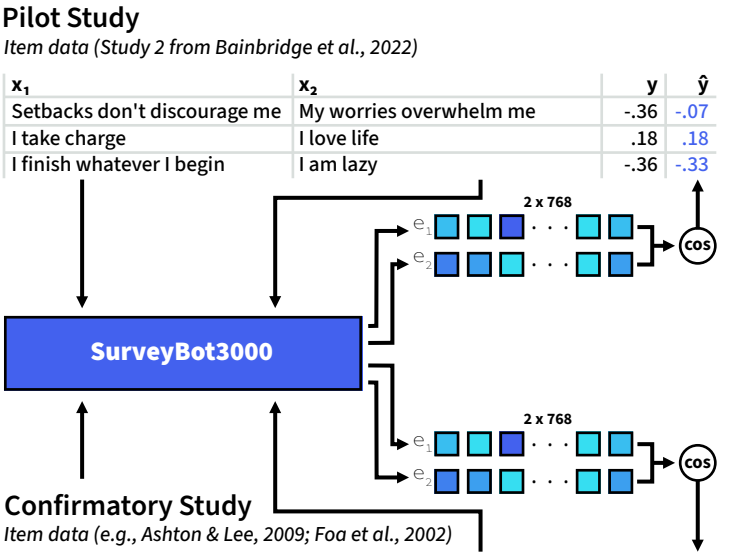
a) Pretraining — Base Model (SBERT)



e. g. SNLI Dataset (Bowman et al., 2015)

x_1	x_2	y	ŷ
cat ruined my life	i love my cat	contradiction	contradiction
dog ate my manuscript	pet interfered with work	entailment	entailment
manatee took my money	the horse was hoarse	neutral	contradiction

c) Validation

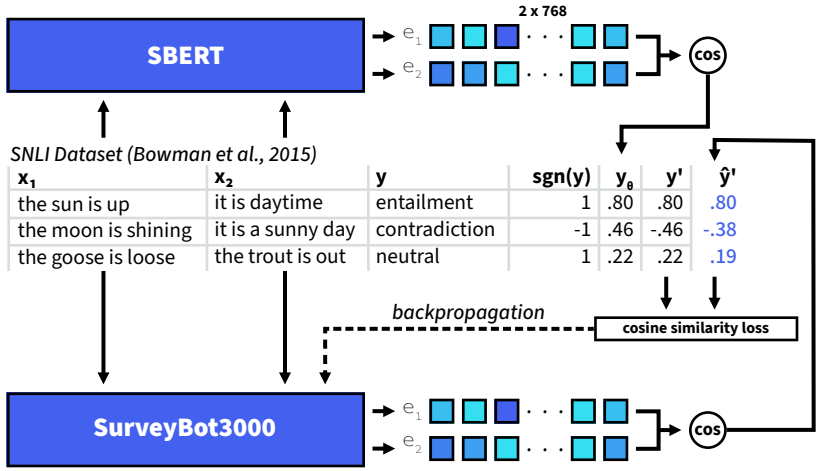


x_1	x_2	y	ŷ
Setbacks don't discourage me	My worries overwhelm me	-.36	-.07
I take charge	I love life	.18	.18
I finish whatever I begin	I am lazy	-.36	-.33

x_1	x_2	y	ŷ
I felt lonely	I lack companionship	*.69*	.52
My job inspires me	I am an outgoing person	*.34*	.23
Completing surveys is a waste of time	I was happy	*-.14*	-.18

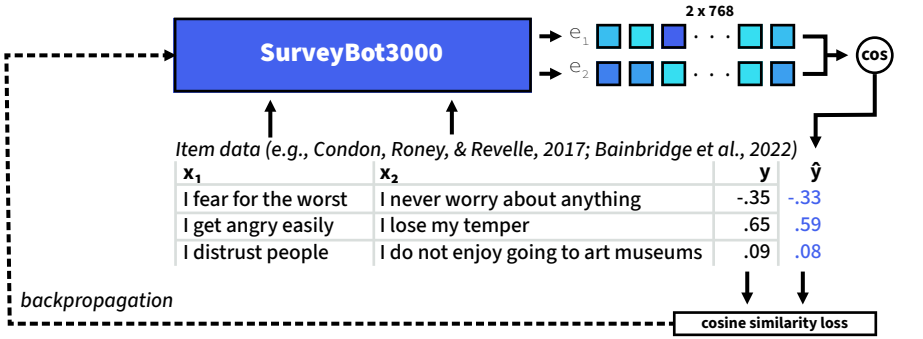
b) Finetuning — SurveyBot3000

Step I: Polarity Calibration



x_1	x_2	y	sgn(y)	y_θ	y'	ŷ'
the sun is up	it is daytime	entailment	1	.80	.80	.80
the moon is shining	it is a sunny day	contradiction	-1	.46	-.46	-.38
the goose is loose	the trout is out	neutral	1	.22	.22	.19

Step II: Domain Adaptation



x_1	x_2	y	ŷ
I fear for the worst	I never worry about anything	-.35	-.33
I get angry easily	I lose my temper	.65	.59
I distrust people	I do not enjoy going to art museums	.09	.08